

RENUKA PRASAD B.S

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GitHub — LinkedIn — Portfolio

Professional Summary

Full Stack & AI Engineer with hands-on experience building scalable web applications, AI-powered systems, and automation workflows across edtech and fintech platforms. Worked on production-grade products at CL Infotech including MindMatrix and TechCovate, contributing to frontend systems, backend APIs, cloud deployment, RAG pipelines, semantic search, analytics automation, and database optimization. Skilled in React.js, Node.js, Fastify, Python, PostgreSQL, MongoDB, Google Cloud Platform, and LLM integrations.

Education

M.S. Ramaiah Institute of Technology

B.E. in Electronics & Communication Engineering — CGPA: 7.3

– Minor Degree in Data Analytics

Bangalore, India

2022 – 2026

Vidya Nidhi Independent PU College

PCMCs — Percentage: 84.33%

Tumkur, India

2020 – 2022

K.V.I.R.S

ICSE — Percentage: 91.6%

Kunigal, India

2010 – 2020

Technical Skills

Languages: Python, JavaScript, TypeScript, SQL

Frontend: React.js, Next.js, Tailwind CSS

Backend: Node.js, Express.js, Fastify, Authentication, Caching, REST APIs

Databases: PostgreSQL, MongoDB

AI Engineering: Prompt Engineering, RAG Pipelines, LLM Integrations, Semantic Search, Vector Databases, Embeddings

Cloud & Tools: Google Cloud Platform (GCP), NGINX, Caddy, Docker, Git, Postman

Experience

Software Engineer

CL Infotech — MindMatrix & TechCovate

Jan 2026 – Present

Bangalore, India

- Built and maintained scalable full-stack modules for edtech and brokerage platforms handling operational and analytics workflows.
- Developed responsive frontend interfaces using React.js and integrated backend services using Node.js, Express.js, and Fastify.
- Designed and optimized SQL/NoSQL database operations for analytics, reporting, and application performance.
- Contributed to AI-powered systems using Retrieval-Augmented Generation (RAG), embeddings, semantic search, and LLM workflows.
- Automated reporting and data extraction pipelines using Python, improving operational efficiency and reducing manual work.
- Worked on deployment, debugging, testing, and agile collaboration within production-focused engineering teams.

MERN Stack Developer Intern

Dhee Coding Lab

June 2025 – Dec 2025

Bangalore, India

- Designed modular microservice architectures to improve scalability and maintainability of applications.
- Developed secure RESTful APIs with JWT authentication and backend validation for client-facing applications.

Projects

AI-Powered RAG Chatbot System — Python, Vector DBs, REST APIs

- Built a Retrieval-Augmented Generation chatbot with semantic search, embeddings, and LLM integration for context-aware responses.
- Designed scalable backend APIs and database workflows for intelligent conversational systems.

Risk Forge — Python, Machine Learning, Streamlit

- Developed a stock portfolio analysis platform using machine learning models and Monte Carlo simulations to predict market trends and investment risks.
- Built interactive dashboards for portfolio visualization and risk assessment.

Attendance & Data Automation System — Python, Pandas, Excel Automation

- Automated extraction, processing, and report generation workflows for attendance analytics using Python-based automation pipelines.

Bulk SMTP Email Automation System — Node.js, SMTP, React.js

- Developed a scalable bulk email automation platform with SMTP integration, campaign management, and automated email delivery workflows.
- Built backend APIs for email queue handling, scheduling, delivery tracking, and dynamic template-based email generation.
- Designed dashboard interfaces for campaign monitoring, analytics, and user management.

Certifications & Achievements

- Solved 200+ coding problems on LeetCode with strong foundations in Data Structures and Algorithms.
- 6× winner in coding competitions conducted by MSRIT and MSRUAS.
- MongoDB Certified — foundational NoSQL database expertise.
- Open-source contributor for THE RESONATE Google Summer of Code initiative using JavaScript, React, and Git workflows.
- Selected as Scholar & Mentee under the FFE Mentorship Program for professional and technical development.